

Cyber Security Task 3 – Secure File Sharing System

Overview

This task demonstrates a secure file sharing system where uploaded files are protected using AES encryption. Files are encrypted before storage and decrypted only when downloaded, ensuring data confidentiality.

Technology Used

- Backend: Python Flask: <https://flask.palletsprojects.com/en/latest/>
- Encryption: AES (PyCryptodome): <https://pycryptodome.readthedocs.io/en/latest/>
- Frontend: HTML / CSS
- Version Control: Git & GitHub

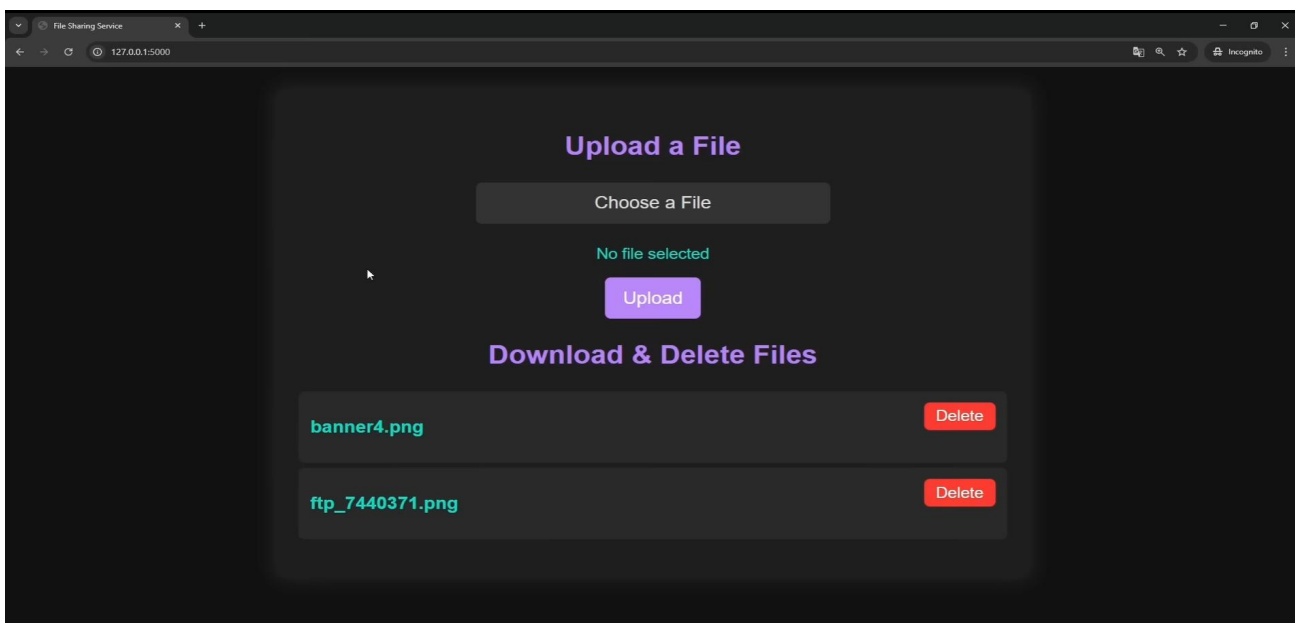
System Workflow

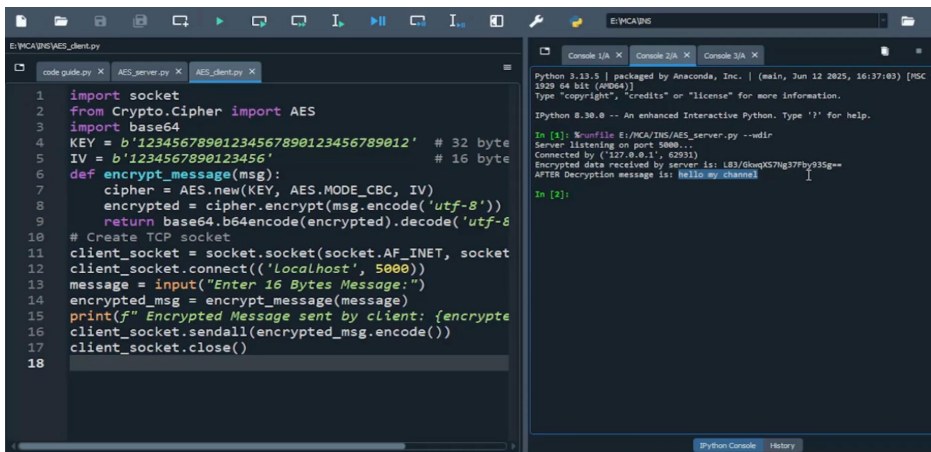
- User uploads a file using the web interface
- File is encrypted using AES
- Encrypted file is stored securely on the server
- On download request, the file is decrypted and sent to the user

Key Security Features

- AES encryption for files at rest
- Secure handling of encryption keys
- File integrity maintained during upload and download
- Restricted access to encrypted files

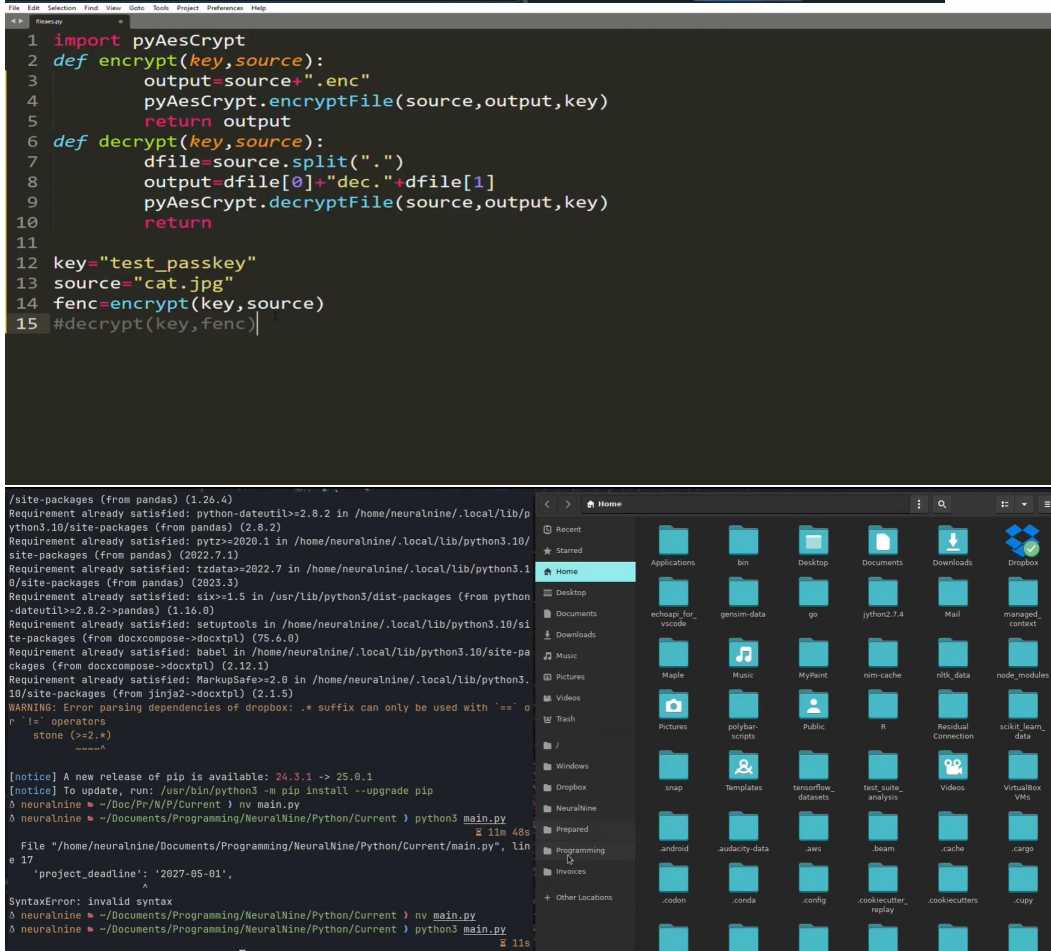
Screenshots





```
1 import socket
2 from Crypto.Cipher import AES
3 import base64
4 KEY = b'12345678901234567890123456789012' # 32 byte
5 IV = b'1234567890123456' # 16 byte
6 def encrypt_message(msg):
7     cipher = AES.new(KEY, AES.MODE_CBC, IV)
8     encrypted = cipher.encrypt(msg.encode('utf-8'))
9     return base64.b64encode(encrypted).decode('utf-8')
10 # Create TCP socket
11 client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
12 client_socket.connect(('localhost', 5000))
13 message = input("Enter 16 Bytes Message:")
14 encrypted_msg = encrypt_message(message)
15 print(f"Encrypted Message sent by client: {encrypted_msg}")
16 client_socket.sendall(encrypted_msg.encode())
17 client_socket.close()
18
```

```
Python 3.11.5 | packaged by Anaconda, Inc. | (main, Jun 12 2025, 16:37:03) [MSC v.1929 64 bit (AMD64)]
Type "copyright", "credits" or "license()" for more information.
Python 3.11.5 -- An enhanced Interactive Python. Type '?' for help.
In [1]: %runfile E:\MCA\INS\AES_server.py --wdir
Server listening on port 5000...
Connected by ("127.0.0.1", 62931)
Encrypted data received by server is: LR3/GkwnXS7Hg377Fb935g==
AFTER Decryption message is: Hello my channel
In [2]:
```



```
1 import pyAesCrypt
2 def encrypt(key, source):
3     output=source+".enc"
4     pyAesCrypt.encryptFile(source,output,key)
5     return output
6 def decrypt(key, source):
7     dfile=source.split(".")
8     output=dfile[0]+".dec."+dfile[1]
9     pyAesCrypt.decryptFile(source,output,key)
10    return
11
12 key="test_passkey"
13 source="cat.jpg"
14 fenc=encrypt(key,source)
15 #decrypt(key,fenc)
```

```
/site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in /home/neuralnine/.local/lib/python3.10/site-packages (from pandas) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /home/neuralnine/.local/lib/python3.10/site-packages (from pandas) (2022.7.1)
Requirement already satisfied: tzdata>=2022.7 in /home/neuralnine/.local/lib/python3.10/site-packages (from pandas) (2023.3)
Requirement already satisfied: six>=1.5 in /usr/lib/python3/dist-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Requirement already satisfied: setuptools in /home/neuralnine/.local/lib/python3.10/site-packages (from docxcompose->docxtpl) (75.6.0)
Requirement already satisfied: babel in /home/neuralnine/.local/lib/python3.10/site-packages (from docxcompose->docxtpl) (2.12.1)
Requirement already satisfied: MarkupSafe>=2.0 in /home/neuralnine/.local/lib/python3.10/site-packages (from jinja2->docxtpl) (2.1.5)
WARNING: Error parsing dependencies of dropbox: .* suffix can only be used with '=' or '!=' operators
stone (>=2.)*
-----
[notice] A new release of pip is available: 24.3.1 -> 25.0.1
[notice] To update, run: /usr/bin/python3 -m pip install --upgrade pip
neuralnine ~ /Doc/PC/N/P/Current > nv main.py
neuralnine ~ /Documents/Programming/NeuralNine/Python/Current > python3 main.py
File "/home/neuralnine/Documents/Programming/NeuralNine/Python/Current/main.py", line 17, in
    'project_deadline': '2027-05-01',
SyntaxError: invalid syntax
neuralnine ~ /Documents/Programming/NeuralNine/Python/Current > nv main.py
neuralnine ~ /Documents/Programming/NeuralNine/Python/Current > python3 main.py
neuralnine ~ /Doc/PC/N/P/Current > |
```

Security Overview

- AES (Advanced Encryption Standard) is used to protect files
- Encryption occurs before saving the file
- Decryption occurs only during file download
- Encryption keys are securely managed on the server

Testing Performed

- Uploaded sample files (PDF, TXT)
- Verified files are stored in encrypted format
- Confirmed successful decryption during download
- Ensured original file content remains unchanged

Conclusion

This project demonstrates basic secure file sharing techniques, encryption implementation, and safe file handling practices used in real-world systems.